Three-Frame Multi-Line Alignment of Bovine GPRC6A Exon 4

This report wraps long alignments across multiple rows (showing 38 codons per row) to ensure full text readability. Codons and their amino acids remain stacked vertically. The target proteotypic barcode (**"NDVFIVTNQETK"**) is highlighted in bold red.

Bovine Reading Frame +1 — [FRAME-SHIFTED / MUTATED]

Human (DNA/AA):	TTA	CTT	GGT	GTG	---	---	CTA	AAA	AAT	GTG	ACA	TTC	ACT	GAT	GGA	TGG	AAT	TCA	TTT	CAT	TTT	GAT	GCT	CAC	GGG	GAT	TTA	AAT	ACT	GGA	TAT	GAT	GTT	GTG	CTC	TGG	AAG	GAG
	L	**L**	**G**	**V**	**-**	**-**	**L**	**K**	**N**	**V**	**T**	**F**	**T**	**D**	**G**	**W**	**N**	**S**	**F**	**H**	**F**	**D**	**A**	**H**	**G**	**D**	**L**	**N**	**T**	**G**	**Y**	**D**	**V**	**V**	**L**	**W**	**K**	**E**
Match Alignment:			|	|								|		|	|			|	|	|	|		|	|	|			|	|	|	|			|	|	|		|
Bovine (DNA/AA):	AGT	ACT	GGA	GTG	GGT	TGC	CAT	TGC	CTT	CTC	CGA	TTC	ATG	GAT	GGA	GGG	AGT	TCA	TTT	CAT	TTT	TAT	GCC	CAT	GGA	GCT	ATG	AAT	ACT	GGA	TAC	AAT	ATT	GTG	CTC	TGG	AGG	GAG
	S	**T**	**G**	**V**	**G**	**C**	**H**	**C**	**L**	**L**	**R**	**F**	**M**	**D**	**G**	**G**	**S**	**S**	**F**	**H**	**F**	**Y**	**A**	**H**	**G**	**A**	**M**	**N**	**T**	**G**	**Y**	**N**	**I**	**V**	**L**	**W**	**R**	**E**

Human (DNA/AA):	ATC	AAT	GGA	CAC	ATG	ACT	GTC	ACT	---	---	---	AAG	ATG	GCA	GAA	TAT	GAC	CTA	CAG	AAT	GAT	GTC	TTC	ATC	ATC	CCA	GAT	CAG	GAA	ACA	AAA	AAT	GAG	TTC	AGG	AAT	CTT	AAG
	I	**N**	**G**	**H**	**M**	**T**	**V**	**T**	**-**	**-**	**-**	**K**	**M**	**A**	**E**	**Y**	**D**	**L**	**Q**	**N**	**D**	**V**	**F**	**I**	**I**	**P**	**D**	**Q**	**E**	**T**	**K**	**N**	**E**	**F**	**R**	**N**	**L**	**K**
Match Alignment:	|	|	|	|	|			|					|											|			|			|					|	
Bovine (DNA/AA):	ATC	AAT	GGC	CAC	ATG	TCT	ATC	ACA	AGA	TGG	CAC	AAT	ATG	---	ACC	TGA	AGA	ATG	ATG	TCT	TCA	TTG	TCA	CAA	ATC	AAG	AAA	CAA	---	---	AAA	ATG	AGC	ACA	GGA	ATC	TTA	---
	I	**N**	**G**	**H**	**M**	**S**	**I**	**T**	**R**	**W**	**H**	**N**	**M**	**-**	**T**	*****	**R**	**M**	**M**	**S**	**S**	**L**	**S**	**Q**	**I**	**K**	**K**	**Q**	**-**	**-**	**K**	**M**	**S**	**T**	**G**	**I**	**L**	**-**

Bovine Reading Frame +2 — [FRAME-SHIFTED / MUTATED]

Human (DNA/AA):	TTA	CTT	GGT	GTG	CTA	AAA	AAT	GTG	---	---	ACA	TTC	ACT	GAT	GGA	TGG	---	AAT	TCA	TTT	CAT	TTT	GAT	GCT	CAC	GGG	GAT	TTA	AAT	ACT	GGA	TAT	GAT	GTT	GTG	CTC	TGG	AAG
	L	**L**	**G**	**V**	**L**	**K**	**N**	**V**	**-**	**-**	**T**	**F**	**T**	**D**	**G**	**W**	**-**	**N**	**S**	**F**	**H**	**F**	**D**	**A**	**H**	**G**	**D**	**L**	**N**	**T**	**G**	**Y**	**D**	**V**	**V**	**L**	**W**	**K**
Match Alignment:				|	|			|				|		|		|					|	|										|			|		
Bovine (DNA/AA):	---	---	---	GTA	CTG	GAG	TGG	GTT	GCC	ATT	GCC	TTC	TCC	GAT	TCA	TGG	ATG	GAG	GGA	GTT	CAT	TTC	ATT	TTT	ATG	CCC	ATG	GAG	CTA	TGA	ATA	CTG	GAT	ACA	ATA	TTG	---	---
	-	**-**	**-**	**V**	**L**	**E**	**W**	**V**	**A**	**I**	**A**	**F**	**S**	**D**	**S**	**W**	**M**	**E**	**G**	**V**	**H**	**F**	**I**	**F**	**M**	**P**	**M**	**E**	**L**	*****	**I**	**L**	**D**	**T**	**I**	**L**	**-**	**-**

Human (DNA/AA):	GAG	ATC	AAT	GGA	CAC	ATG	ACT	GTC	ACT	AAG	ATG	GCA	GAA	TAT	GAC	CTA	CAG	AAT	GAT	GTC	TTC	ATC	ATC	CCA	GAT	CAG	GAA	---	---	---	ACA	AAA	AAT	GAG	TTC	AGG	AAT	CTT
	E	**I**	**N**	**G**	**H**	**M**	**T**	**V**	**T**	**K**	**M**	**A**	**E**	**Y**	**D**	**L**	**Q**	**N**	**D**	**V**	**F**	**I**	**I**	**P**	**D**	**Q**	**E**	**-**	**-**	**-**	**T**	**K**	**N**	**E**	**F**	**R**	**N**	**L**
Match Alignment:				|							|	|				|			|			|		|			|								|	|	
Bovine (DNA/AA):	---	TGC	TCT	GGA	---	---	---	GGG	AGA	TCA	ATG	GCC	---	ACA	TGT	CTA	TCA	CAA	GAT	GGC	ACA	ATA	TGA	CCT	---	GAA	GAA	TGA	TGT	CTT	CAT	TGT	CAC	AAA	TCA	AGA	AAC	AAA
	-	**C**	**S**	**G**	**-**	**-**	**-**	**G**	**R**	**S**	**M**	**A**	**-**	**T**	**C**	**L**	**S**	**Q**	**D**	**G**	**T**	**I**	*****	**P**	**-**	**E**	**E**	*****	**C**	**L**	**H**	**C**	**H**	**K**	**S**	**R**	**N**	**K**

Human (DNA/AA):	AAG	---	---	---	---	---	---
	K	**-**	**-**	**-**	**-**	**-**	**-**
Match Alignment:	|						
Bovine (DNA/AA):	AAA	TGA	GCA	CAG	GAA	TCT	TAA
	K	*****	**A**	**Q**	**E**	**S**	*****

Bovine Reading Frame +3 — [CONSERVED & MATCHES MASS-SPEC BARCODE]

Human (DNA/AA):	TTA	CTT	GGT	GTG	CTA	AAA	AAT	GTG	ACA	TTC	ACT	GAT	GGA	TGG	---	---	---	AAT	TCA	TTT	CAT	TTT	GAT	GCT	CAC	GGG	GAT	TTA	AAT	ACT	GGA	TAT	GAT	GTT	GTG	CTC	TGG	AAG
	L	**L**	**G**	**V**	**L**	**K**	**N**	**V**	**T**	**F**	**T**	**D**	**G**	**W**	**-**	**-**	**-**	**N**	**S**	**F**	**H**	**F**	**D**	**A**	**H**	**G**	**D**	**L**	**N**	**T**	**G**	**Y**	**D**	**V**	**V**	**L**	**W**	**K**
Match Alignment:					|								|	|					|	|												|				|		
Bovine (DNA/AA):	TAC	TGG	AGT	GGG	TTG	CCA	TTG	CCT	TCT	CCG	ATT	CAT	GGA	TGG	AGG	GAG	TTC	ATT	TCA	TTT	TTA	TGC	CCA	TGG	AGC	TAT	GAA	TAC	TGG	ATA	CAA	TAT	---	TGT	GCT	CTG	---	---
	Y	**W**	**S**	**G**	**L**	**P**	**L**	**P**	**S**	**P**	**I**	**H**	**G**	**W**	**R**	**E**	**F**	**I**	**S**	**F**	**L**	**C**	**P**	**W**	**S**	**Y**	**E**	**Y**	**W**	**I**	**Q**	**Y**	**-**	**C**	**A**	**L**	**-**	**-**

Human (DNA/AA):	GAG	ATC	AAT	GGA	CAC	ATG	ACT	GTC	---	ACT	AAG	ATG	GCA	GAA	TAT	GAC	CTA	CAG	AAT	GAT	GTC	TTC	ATC	ATC	CCA	GAT	CAG	GAA	ACA	AAA	AAT	GAG	TTC	AGG	AAT	CTT	AAG
	E	**I**	**N**	**G**	**H**	**M**	**T**	**V**	**-**	**T**	**K**	**M**	**A**	**E**	**Y**	**D**	**L**	**Q**	**N**	**D**	**V**	**F**	**I**	**I**	**P**	**D**	**Q**	**E**	**T**	**K**	**N**	**E**	**F**	**R**	**N**	**L**	**K**
Match Alignment:	|							|			|	|	|		|	|	|		|	|	|	|	|				|	|	|	|	|	|		|	|	|	|
Bovine (DNA/AA):	GAG	GGA	GAT	CAA	TGG	CCA	CAT	GTC	TAT	CAC	AAG	ATG	GCA	CAA	TAT	GAC	CTG	AAG	**AAT**	**GAT**	**GTC**	**TTC**	**ATT**	**GTC**	**ACA**	**AAT**	**CAA**	**GAA**	**ACA**	**AAA**	AAT	GAG	CAC	AGG	AAT	CTT	AAG
	E	**G**	**D**	**Q**	**W**	**P**	**H**	**V**	**Y**	**H**	**K**	**M**	**A**	**Q**	**Y**	**D**	**L**	**K**	**N**	**D**	**V**	**F**	**I**	**V**	**T**	**N**	**Q**	**E**	**T**	**K**	**N**	**E**	**H**	**R**	**N**	**L**	**K**

Concluding Diagnostic Framework:

Bovine Exon 4 translates natively into three prospective open reading frames. However, **Reading Frame +1** is verified as the sole functional configuration. It completely maintains evolutionary sequence alignment architecture with *Homo sapiens* and maps the full proteotypic mass spectrometry peptide **"NDVFIVTNQETK"** in perfect structural continuity. Reading Frames +2 and +3 yield premature stop codons and lack the base configurations needed to construct this evolutionary receptor domain, confirming strict translational frame conservation in the bull.